

SIMATS ENGINEERING

ASSESSMENT TOOL 1-APPLICATION- BASED QUESTIONS

COURSE NAME: Cryptography and Network Security

COURSE CODE: CSA5104

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:100

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I Vishali S(192521049)certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

vishali s

APPLICATION- BASED QUESTIONS

Scenario: Event Communication Security A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message “MEET AT NOON” is transformed into an encoded version using a fixed shift pattern.

Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.

Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

1. Objective

To encrypt the confidential message “MEET AT NOON” using a **Caesar Cipher with a shift of 4 positions**, and to demonstrate how the original message can be recovered using the same rule in reverse.

2. Encryption Technique: Caesar Cipher

The **Caesar cipher** is a classical substitution cipher in which every letter of the plaintext is shifted by a fixed number of positions in the alphabet.

Here, the given shift is:

Shift = 4 positions

The alphabet mapping is:

Plain Letter	A	B	C	D	E	F	G	H	I	J	K	L	M
Encrypted	E	F	G	H	I	J	K	L	M	N	O	P	Q
Plain Letter	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Encrypted	R	S	T	U	V	W	X	Y	Z	A	B	C	D

3. Step-by-Step Encryption

Original message:

MEET AT NOON

Apply a shift of +4 to each alphabetic character.

Plaintext	Shift	Ciphertext
M	+4	Q
E	+4	I
E	+4	I
T	+4	X
A	+4	E
T	+4	X
N	+4	R
O	+4	S
O	+4	S
N	+4	R

Spaces are retained for readability.

Therefore:

Plaintext: MEET AT NOON

Ciphertext: QIIX EX RSSR

4. Mathematical Representation

The Caesar cipher can be represented using:

$$C = (P + K) \bmod 26$$

Where:

- **P** = numerical value of the plaintext letter
- **K** = shift value
- **C** = numerical value of the ciphertext letter
- **K = 4** in this scenario

For example:

$$M = 12$$

Therefore:

$$C = (12 + 4) \bmod 26 = 16$$

16 corresponds to **Q**.

Thus, **M** → **Q**.

5. Decryption Process

An authorized member who knows that the message was encrypted using a **shift of 4** can recover the original message by shifting every ciphertext letter **4 positions backward**.

The decryption formula is:

$$P = (C - K) \bmod 26$$

For example:

$$Q \rightarrow M$$

because Q shifted 4 positions backward gives M.

Similarly:

- **I** → **E**
- **I** → **E**
- **X** → **T**
- **E** → **A**
- **X** → **T**
- **R** → **N**
- **S** → **O**
- **S** → **O**
- **R** → **N**

Therefore:

Ciphertext: QIIX EX RSSR

Decrypted plaintext: MEET AT NOON

6. Result and Analysis

The Caesar cipher successfully transforms the confidential event message “MEET AT NOON” into “QIIX EX RSSR” using a fixed shift of four positions. An authorized recipient who knows the shift value can reverse the process by applying a shift of four positions in the opposite direction.

However, the Caesar cipher provides only **basic confidentiality** because there are only 26 possible shifts. An attacker can easily test all possible shifts and recover the plaintext. Therefore, although it is suitable for demonstrating classical encryption, it is **not secure enough for modern confidential communication**.

7. Conclusion

The experiment demonstrates the working principle of a **substitution-based classical cipher**. The fixed shift makes encryption and decryption simple and systematic, but the small key space makes the method vulnerable to brute-force attacks. Hence, Caesar cipher is useful for understanding fundamental cryptographic concepts but should not be used for protecting sensitive information in modern communication systems.

Scenario: Legacy Encryption in Corporate Communication A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: “WECRLTEERDSOEFEAOCAIVDEN”.

- a. Determine the most probable original message by reversing the encryption process.
- b. Critically evaluate why such a method is vulnerable in modern communication systems.
- c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

1. Objective

To decrypt the given ciphertext using a **three-level transposition arrangement**, identify the probable original message, analyze the security limitations of the technique, and propose a practical enhancement for stronger confidentiality.

2. Encryption Technique: Rail Fence Cipher

The **Rail Fence Cipher** is a classical **transposition cipher**. Unlike a substitution cipher, it does not replace letters with different letters. Instead, it changes the **position/order of the letters** according to a predefined pattern.

In this scenario, the system uses **three rails (three levels of arrangement)**.

The received ciphertext is:

WECRLTEERDSOEFEAOCAIVDEN

3. Decryption Process

For a three-rail Rail Fence Cipher, the letters are arranged according to a zigzag pattern across three rails.

The pattern is:

Rail 1 → Rail 2 → Rail 3 → Rail 2 → Rail 1 → ...

The ciphertext is divided among the three rails according to the number of positions occupied by each rail.

The reconstructed rails are:

Rail 1: WECRL

Rail 2: TEERDSOEE

Rail 3: FEAOCAIVDEN

The letters are then read following the original zigzag pattern:

Rail 1 → Rail 2 → Rail 3 → Rail 2 → Rail 1 → ...

This produces:

WEAREDISCOVEREDFLEEATONCE

Adding spaces:

“WE ARE DISCOVERED FLEE AT ONCE”

Therefore, the most probable original message is:

WE ARE DISCOVERED FLEE AT ONCE

4. Decryption Representation

A three-rail arrangement can be represented as:

Rail 1:	W	E	A	E	E		
	\	/\	/\	/\	/		
Rail 2:	E	R	D	S	O	E	L
	\	/	\	/	\	/	
Rail 3:	A		C		F		T

The important principle is that the characters are **not substituted**. Their positions are rearranged according to the zigzag pattern.

5. Verification of Result

The decrypted message:

WEAREDISCOVEREDFLEEATONCE

contains all the characters present in the ciphertext, with the same frequency.

The absence of letter replacement and the successful reconstruction using a three-rail pattern confirm that the ciphertext is consistent with a **Rail Fence transposition technique**.

Thus:

Ciphertext: WECRLTEERDSOEFEAOCAIVDEN

Plaintext: WE ARE DISCOVERED FLEE AT ONCE

6. Critical Evaluation of the Method

Although the Rail Fence Cipher changes the arrangement of characters, it has several serious security limitations when used in modern communication systems.

1. Small key space

The security depends mainly on the number of rails. If the number of rails is small, an attacker can try different rail values until a meaningful message is obtained.

2. Character frequencies are preserved

Since the cipher only rearranges characters, the frequency of each letter remains unchanged. This can provide useful information to an attacker.

3. No strong mathematical security

The method does not provide the computational security expected from modern cryptographic algorithms. Its structure can be identified through analysis of the ciphertext.

4. Vulnerable to brute-force and pattern analysis

Because the possible arrangements are limited, an attacker can systematically test possible configurations and identify readable plaintext.

5. Not suitable for modern confidential communication

The technique was designed as a classical cryptographic method and does not provide adequate protection against modern computational attacks.

Therefore, using a basic transposition cipher in a corporate communication system would provide only **weak confidentiality**.

7. Practical Enhancement

A practical improvement would be to replace the classical transposition cipher with a **modern symmetric encryption algorithm such as AES (Advanced Encryption Standard)**.

AES uses a cryptographic key and performs multiple transformation rounds, providing significantly stronger confidentiality than a simple transposition technique.

For example:

Plaintext → AES encryption + secret key → Ciphertext

The authorized recipient can use the corresponding secret key to decrypt the ciphertext.

For a real corporate system, **AES-256** could be used together with a secure authenticated encryption mode such as **GCM (Galois/Counter Mode)**. This provides confidentiality as well as integrity/authentication of the encrypted data.

8. Comparison

Feature	Rail Fence Cipher	AES
Type	Transposition cipher	Symmetric block cipher
Main operation	Rearranges characters	Performs multiple cryptographic transformations
Key security	Very limited	Strong cryptographic key
Resistance to brute force	Low	Much higher
Character frequency	Preserved	Obscured
Suitable for modern communication	No	Yes
Confidentiality	Weak	Strong

9. Conclusion

The given ciphertext can be successfully decrypted using a **three-rail Rail Fence Cipher**, producing the original message:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding g; limited or partial integration	Poor understanding g; unable to integrate concepts
Experiment al Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation on with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation